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Papers 847 (concl.), 848 (start)

Arthur Cayley

## 847 (concluded). On the Theory of Seminvariants.

[Printed p. 356]

To obtain the biliteral terms in the proper form 1° we write therein  $1 = 0$ , 2° we write  $\beta = 0$ , changing  $\gamma$  into  $\beta$ , 3° we write  $a = 0$ , changing  $\beta, \gamma$  into  $a, \beta$ ; we thus obtain

$$(a - \beta)^{24}(a + \beta)a\beta, \quad a^{24}(a - \beta)(-\beta), \quad a^{24}(a - 2\beta)(-\beta)(a - \beta);$$

viz. the second and third terms are identical; the first term requires, however, the factor 2 (for any term  $a^p\beta^q$  is accompanied by a corresponding term  $a^q\beta^p$ ), so that, omitting this factor throughout, the terms are

$$(a - \beta)^{24}(a + \beta)a\beta - a^{24}(a - 2\beta)\beta(a - \beta),$$

or, putting therein  $a = 1$ , the terms are

$$(1 - \beta)^{24}(1 + \beta)\beta - (1 - 2\beta)(1 - \beta)\beta.$$

Similarly from

$$Z = (a - \beta)^{22}(a - \beta - 2\gamma)(a - \gamma)^2(\beta - \gamma)^2,$$

we have

$$(1 - \beta)^{22}(1 + \beta)\beta^2 + (1 - 2\beta)(1 - \beta)^2\beta^2,$$

and the like as regards  $W$  and  $T$ . The result thus is

$$(1 + \beta)(1 - \beta)^{24} + \lambda(1 - \beta)^{22}\beta + \mu(1 - \beta)^{20}\beta^2 + \nu(1 - \beta)^{18}\beta^3 - (1 - 2\beta)(1 - \beta)\beta\{1 - \lambda\beta(1 - \beta) + \mu\beta^2(1 - \beta)^2 - \nu\beta^3(1 - \beta)^3\} = 0,$$

or, as this may be written,

$$\frac{(1 + \beta)(1 - \beta)^{24}}{1 - 2\beta} \left\{ 1 + \lambda \frac{\beta}{(1 - \beta)^2} + \mu \frac{\beta^2}{(1 - \beta)^4} + \nu \frac{\beta^3}{(1 - \beta)^6} + \dots \right\} \\ - 1 + \lambda(\beta - \beta^2) - \mu(\beta - \beta^2)^2 + \nu(\beta - \beta^2)^3 - \dots = 0,$$

viz. each side is to be expanded in ascending powers of  $\beta$ , and the coefficients  $\lambda, \mu, \nu$  are then to be determined so that as many terms as possible shall vanish.

Expanding, we have

$$(1 - 20\beta + 190\beta^2 - 1138\beta^3 + 4808\beta^4 - 15178\beta^5 + \dots) \\ \times \{1 + \beta\lambda + \beta^2(2\lambda + \mu) + \beta^3(3\lambda + 4\mu + \nu) + \beta^4(4\lambda + 10\mu + 6\nu) + \beta^5(5\lambda + 20\mu + 21\nu) + \dots\} \\ - 1 + \beta\lambda + \beta^2(-\lambda - \mu) + \beta^3(2\mu + \nu) + \beta^4(-3\nu) + \beta^5 \cdot 3\nu + \dots = 0,$$

or finally, as far as  $\beta^5$ , the equation is

$$0 = 0 + \beta(2\lambda - 20) + \beta^2(-19\lambda + 190) + \beta^3(153\lambda - 14\mu + 2\nu - 1138) + \beta^4(-814\lambda + 119\mu - 17\nu + 4808) + \beta^5(3027\lambda - 558\mu + 94\nu - 15178).$$

We have in the first place  $\lambda = 10$ , which makes the coefficient of  $\beta^2$  to be  $= 0$ ; and we then have

$$\begin{aligned} -14\mu + 2\nu + 392 &= 0, & \text{or say } 7\mu - \nu - 196 &= 0, \\ 119\mu - 17\nu - 3332 &= 0, & 119\mu - 17\nu - 3332 &= 0, \\ -558\mu + 94\nu + 15092 &= 0, & 272\mu - 47\nu - 7546 &= 0, \end{aligned}$$

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where the second equation is equivalent to the first: the three equations give  $\mu = \frac{273}{8}$ ,  $\nu = \frac{231}{4}$ . I have since found that proceeding one step further, that is, to  $\beta^6$ , the coefficient is  $-8310\lambda + 1791\mu - 363\nu + 36942$ , viz. putting this  $= 0$ , and for  $\lambda$  substituting its value  $= 10$ , we have  $597\mu - 121\nu = 15386$ , an equation which is in fact satisfied by the values just obtained for  $\lambda$  and  $\mu$ . For the function  $(Y, Z, W)$  we have the same equations with  $\nu = 28$  and for the function  $(Y, Z)$  the same equations with  $\mu = 0$ ,  $\nu = 0$ ; and therefore  $\lambda = 10$ . The linear combinations thus are

$$\begin{aligned} X &= a_2c + ba^2, \\ Y &= a_{19}(c + d^2) + a_2(c^2 - b^2), \\ Y + 10Z &= a_{19}(e + c^2) + a_{20}, \\ Y + 10Z + 28W &= a_{21}(g + hk^2 + d^2), \\ 25Y + 250Z + 823W + 931T &= a_{19}(i + k + j^3), \end{aligned}$$

where remark that in  $Y + 10Z$ , by means of the coefficient 10, we make to disappear the two terms  $a_{20}c$ ,  $a_{20}d$ ; in the next function, by means of the coefficients 10 and 28, the four terms  $a_{21}c$ ,  $a_{21}d$ ,  $a_{21}e$ ,  $a_{21}f$ ; and in the last function, by means of the three coefficients, the six terms  $a_{19}c$ ,  $a_{19}d$ ,  $a_{19}e$ ,  $a_{19}f$ ,  $a_{21}g$  and  $a_{21}h$ .

It is possible that a larger number of terms will disappear; but if this is so, the general form shown by the combinations on p. 353 will require modification.

The investigation applies without alteration to any odd weight  $\omega$  whatever; the condition for the determination of the coefficients  $\lambda$ ,  $\mu$ ,  $\nu$ , ... is obviously

$$\frac{(1+\beta)(1-\beta)^{\omega-1}}{1-2\beta} \left\{ 1 + \lambda \frac{\beta}{(1-\beta)^2} + \mu \frac{\beta^2}{(1-\beta)^4} + \nu \frac{\beta^3}{(1-\beta)^6} + \dots \right\} - 1 + \lambda(\beta - \beta^2) - \mu(\beta - \beta^2)^2 + \nu(\beta - \beta^2)^3 - \dots = 0.$$

For the case of an even weight  $\omega$ , the series of functions is

$$\begin{aligned} X &= (a - \beta)^\omega, \\ Y &= (a - \beta)^{\omega-2}(a - \gamma)^2(\beta - \gamma), \\ Z &= (a - \beta)^{\omega-4}(a - \gamma)^2(\beta - \gamma)^2, \end{aligned}$$

and the condition for the determination of the coefficients  $\lambda$ ,  $\mu$ ,  $\nu$ , ... is in like manner found to be

$$(1-\beta)^{\omega-1} \left\{ 1 + \lambda \frac{\beta}{(1-\beta)^2} + \mu \frac{\beta^2}{(1-\beta)^4} + \nu \frac{\beta^3}{(1-\beta)^6} + \dots \right\} - 1 + \lambda(\beta - \beta^2) - \mu(\beta - \beta^2)^2 + \nu(\beta - \beta^2)^3 + \dots = 0,$$

which is to be understood as explained above.

## 848. On the Transformation of the Double-Theta Functions.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xxi. (1886), pp. 142–178.]

[Printed p. 358]

I propose to reproduce Hermite's Memoir "Sur la théorie de la transformation des fonctions Abéliennes," *Comptes Rendus*, t. XL. (1855), pp. 249, ..., 784, with some changes of notation and developments. Hermite's functions are even or odd according as we have  $\mu q + \nu p$  even or odd: viz. his characteristic is  $\begin{pmatrix} \mu, \nu \\ q, p \end{pmatrix}$ , or the letters  $p$ ,  $q$ , are misplaced; I write, therefore,  $r$  instead of  $p$ , so as to have the characteristic  $\begin{pmatrix} \mu, \nu \\ q, r \end{pmatrix}$ ;

and then for symmetry it is necessary to interchange the suffixes 2, 3 and the letters  $c, d$ ; the invariant function of the periods, instead of being as with him

$$\omega_0 v_1 - \omega_1 v_0 + \omega_3 v_2 - \omega_2 v_3,$$

must be taken to be

$$\omega_0 v_1 - \omega_1 v_0 + \omega_2 v_3 - \omega_3 v_2.$$

Moreover, I write  $A, B$  for his  $G, G'$ , so as, instead of

$$(G, H, G' \mid x, y)^2,$$

to have in the expressions of the theta-functions the quadric function  $(A, H, B \mid x, y)^2$ ; and I alter the arrangement of the memoir so as to separate more completely the preliminary theory from the theory of the transformation.

GENERAL THEORY. Art. Nos. 1 to 21 (several sub-headings).

*The functions  $\Pi\{K, \text{indef. or def.}\}$ .*

1. Consider a function

$$\Pi\left(\begin{smallmatrix} \mu, \nu \\ q, r \end{smallmatrix}\right)(x, y)(A, H, B)\{K, \text{indef. or def.}\},$$

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having: the characteristic  $\left(\begin{smallmatrix} \mu, \nu \\ q, r \end{smallmatrix}\right)$ , where the characters  $\mu, \nu, q, r$  are positive or negative integers, which may be taken to have each of them only the values  $(0, 1)$  at pleasure, so that the number of functions is  $2^4 = 16$ ;

the arguments  $(x, y)$ ;

the parameters, or conjoint quarter-periods,  $(A, H, B)$ ;

and the potency  $K$ , a positive integer;

and which is either indefinite or definite, as will be explained; the function, moreover, contains linearly certain arbitrary constants, the number of them depending on the value of  $K$ , as will be explained.

The function may be written  $\Pi(x, y)$  or in any other less abbreviated form which may be convenient.

2. The function  $\Pi(x, y)\{K \text{ indef.}\}$  is defined by the following four equations:

$$\Pi(x + 1, y) = (-)^\mu \Pi(x, y),$$

$$\Pi(x, y + 1) = (-)^\nu \Pi(x, y),$$

$$\Pi(x + A, y + H) = (-)^q \Pi(x, y) \exp. -2\pi K(2x + A),$$

$$\Pi(x + H, y + B) = (-)^r \Pi(x, y) \exp. -2\pi K(2y + B),$$

and the function  $\Pi(x, y)\{K \text{ def.}\}$  by the same equations, together with the following fifth equation,

$$\Pi(-x, -y) = (-)^{\mu q + \nu r} \Pi(x, y);$$

viz. the definite function is an even function or else an odd function of the arguments according as  $\mu q + \nu r$  is even or odd. We may call  $\mu q + \nu r$  the index; and the function is then even or odd according as the index is even or odd.

It is perhaps worth noticing that it would be allowable to define a function  $\Pi(x, y)\{K, \text{skew def.}\}$ , by the corresponding relation

$$\Pi(-x, -y) = -(-)^{\mu q + \nu r} \Pi(x, y),$$

but I do not propose here to develop this notion.

3. The four equations give rise to the following one,

$$\Pi(x + a_0 + Aa_2 + Ha_3, y + a_1 + Ha_2 + Ba_3) = (-)^{\mu a_0 + \nu a_1 + q a_2 + r a_3} \Pi(x, y) \times \exp. -2\pi K\{2a_2x + 2a_3y + (A, H, B \mid a_2, a_3)^2\},$$

where  $a_0, a_1, a_2, a_3$  are any positive or negative integers (zero not excluded), and which single equation, in fact, includes the preceding four equations.

4. In regard to the parameters it is to be observed that, if  $A = A_0 + ia$ ,  $H = H_0 + i\eta$ ,  $B = B_0 + i\beta$ , we must have  $(a, \eta, \beta)$  a determinate positive quadratic form; viz. this is the necessary and sufficient condition for the convergence of the series for the development of the function.

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5. The number of arbitrary constants is for the indefinite function  $= K^2$ ; but for the definite function it is, when  $K$  is odd,  $= \frac{1}{2}(K^2 + 1)$ ; and when  $K$  is even, it is  $= \frac{1}{2}K^2$ , except in the case of a characteristic  $\binom{0,0}{q,r}$ , when it is  $= \frac{1}{2}(K^2 + 4)$ .

In particular, for  $K = 1$ , there is only a single arbitrary constant, which is a mere factor of the function; taking it to be  $= 1$ , as presently explained, we have the 16 theta-functions.

6. The function  $\Pi(x, y)$  is developed in a series of exponentials, in the form

$$\Pi(x, y) = \Sigma(-)^{mq+nr} A_{m,n} \exp. i\pi \left\{ (2m + \mu)x + (2n + \nu)y + \frac{1}{4K}(A, H, B \mid 2m + \mu, 2n + \nu)^2 \right\},$$

where  $m$  and  $n$  have each of them all positive and negative integer values (zero not excluded) from  $-\infty$  to  $\infty$ . In fact, substituting this series in the four equations, they are all of them satisfied if only

$$A_{m+K,n} = A_{m,n}; \quad A_{m,n+K} = A_{m,n}.$$

Consequently the following  $K^2$  coefficients remain arbitrary, viz. those with the suffixes

$$0, 1, \dots, K-1;$$

and we have for  $\Pi(x, y)$  a sum of  $K^2$  terms, each a determinate series multiplied into one of the arbitrary coefficients  $A_{0,0}$ ,  $A_{0,1}$ , &c. The indefinite function thus contains, as already mentioned,  $K^2$  arbitrary constants.

7. Substituting in the fifth equation, we have for the definite function the further condition

$$A_{-m,-n} = A_{m,n},$$

which it is clear will be satisfied generally if only it is satisfied by the coefficients in the foregoing set of  $K^2$  coefficients.

8. In the case  $K$  odd, we thus reduce the number of arbitrary coefficients to  $\frac{1}{2}(K^2 + 1)$ ; the mode in which this takes place is best seen by an example. Suppose  $K = 3$ , so that  $A_{m+3,n} = A_{m,n}$ ;  $A_{m,n+3} = A_{m,n}$ . For the coefficients of the indefinite function, the suffixes are

$$\begin{array}{ccc} 00, & 01, & 02, \\ 10, & 11, & 12, \\ 20, & 21, & 22. \end{array}$$

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And if we suppose  $\mu = 0$ ,  $\nu = 1$ , then the new condition is  $A_{-m,-n-1} = A_{m,n}$ , viz. writing down only the suffixes, we thus obtain

$$\begin{array}{lll} 0, 0 = 0, -1, & 1, 0 = -1, -1, & 2, 0 = -2, -1, \\ 0, 1 = 0, -2, & 1, 1 = -1, -2, & 2, 1 = -2, -2, \\ 0, 2 = 0, -3, & 1, 2 = -1, -3, & 2, 2 = -2, -3, \end{array}$$

that is,

$$\begin{array}{lll} 0, 0 = 0, 2, & 1, 0 = 2, 2, & 2, 0 = 1, 2, \\ 0, 1 = 0, 1, & 1, 1 = 2, 1, & 2, 1 = 1, 1, \\ 0, 2 = 0, 0, & 1, 2 = 2, 0, & 2, 2 = 1, 0, \end{array}$$

viz. one of these equations  $0, 1 = 0, 1$  is an identity, but the other equations occur each twice; or we have four equations, each of them an equality between two out of the remaining 8 coefficients; the number of arbitrary coefficients is thus  $1 + (9 - 1)/2 = 5$ ,  $= \frac{1}{2}(K^2 + 1)$ .

9. When  $K$  is even, it is necessary to distinguish between the case  $(\mu, \nu) = (0, 0)$  and the remaining three cases  $(\mu, \nu) = (1, 0)$ ,  $(0, 1)$  or  $(1, 1)$ . In the former case, the relation between the coefficients is

$A_{-m,-n} = A_{m,n}$ ; there are four identities,  $0, 0 = 0, 0; 0, \frac{1}{2}K = 0, \frac{1}{2}K; \frac{1}{2}K, 0 = \frac{1}{2}K, 0; \frac{1}{2}K, \frac{1}{2}K = \frac{1}{2}K, \frac{1}{2}K$ ; and the remaining  $K^2 - 4$  equations occur each twice, that is, we have  $\frac{1}{2}(K^2 - 4)$  equations, each of them an equality between two of the remaining  $K^2 - 4$  coefficients; and we thus have  $\frac{1}{2}K^2$  arbitrary coefficients.

In the latter case, there are no identities and the  $K^2$  equations occur each twice, that is, we have  $\frac{1}{2}K^2$  equations, each of them an equality between two of the  $K^2$  coefficients; and we thus have  $\frac{1}{2}K^2$  arbitrary coefficients.

**10.** Recapitulating, it thus appears that

$$\begin{aligned} &\text{for an indefinite function, the number of coefficients} = K^2; \\ &\text{for a definite function, the number} = \frac{1}{2}(K^2 + 1), \quad K \text{ odd}; \\ &= \frac{1}{2}K^2, \quad K \text{ even, and} \\ &(\mu, \nu) = (1, 0), (0, 1) \text{ or } (1, 1); \\ &= \frac{1}{2}(K^2 + 4), \quad K \text{ even, and} \\ &(\mu, \nu) = (0, 0). \end{aligned}$$

### *The Theta-functions.*

**11.** In the particular case  $K = 1$ , the distinction between the indefinite function and the definite function disappears, and we have instead of  $\Pi(x, y)$ , the theta-functions  $\Theta(x, y)$ , satisfying the four equations

$$\begin{aligned} \Theta(x + 1, y) &= (-)^\mu \Theta(x, y), \\ \Theta(x, y + 1) &= (-)^\nu \Theta(x, y), \\ \Theta(x + A, y + H) &= (-)^q \Theta(x, y) \exp. -i\pi(2x + A), \\ \Theta(x + H, y + B) &= (-)^r \Theta(x, y) \exp. -i\pi(2y + B), \end{aligned}$$

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and the fifth equation

$$\Theta(-x, -y) = (-)^{\mu q + \nu r} \Theta(x, y);$$

as before,  $\mu q + \nu r$  is the index of the function.

The four equations are all of them included in the following one:

$$\Theta(x + a_0 + Aa_2 + Ha_3, y + a_1 + Ha_2 + Ba_3) = (-)^{\mu a_0 + \nu a_1 + q a_2 + r a_3} \Theta(x, y) \times \exp. -i\pi\{2a_2x + 2a_3y + (A, H, B \mid a_2, a_3)^2\},$$

where  $a_0, a_1, a_2, a_3$  are each of them any positive or negative integer, zero not excluded.

Moreover, we take  $A_{0,0} = 1$ , and the value of the function thus is

$$\Theta(x, y) = \Sigma(-)^{mq+nr} \exp. i\pi\{(2m + \mu)x + (2n + \nu)y + \frac{1}{4}(A, H, B \mid 2m + \mu, 2n + \nu)^2\}.$$

**12.** The sum of two characteristics is the characteristic obtained by taking the sums of the component terms or characters,

$$\begin{pmatrix} \mu, \nu \\ q, r \end{pmatrix} + \begin{pmatrix} \mu', \nu' \\ q', r' \end{pmatrix} = \begin{pmatrix} \mu + \mu', \nu + \nu' \\ q + q', r + r' \end{pmatrix};$$

and similarly for any number of characteristics. I use the sign  $=$ , but this properly denotes a congruence, mod 2; and the like as regards the indices.

The sum of two identical characteristics, or generally of any number of characteristics taken each of them any even number of times, is  $\begin{pmatrix} 0, 0 \\ 0, 0 \end{pmatrix}$ . And this characteristic  $\begin{pmatrix} 0, 0 \\ 0, 0 \end{pmatrix}$  may be called the characteristic 0.

It should be observed, that the index of the sum is not in general equal to the sum of the indices. To make it so, we must have, for two characteristics,

$$(\mu + \mu')(q + q') + (\nu + \nu')(r + r') = \mu q + \nu r + \mu' q' + \nu' r',$$

that is,

$$\mu q' + \nu' r + \mu' q + \nu r' = 0;$$

and there is obviously a like formula for the case of more than two characteristics.

Two or more characteristics, such that they have the sum of the indices equal to be the index of the sum, are said to be in “direct relation” or “directly related” to each other. The sum of the indices and the index of the sum may differ by unity; and we then have the inverse relation; but I do not propose to consider this.

**13.** Consider any number  $K$  of theta-functions, of the same arguments and parameters, but with the same or different characteristics. The product of these functions is in general a function  $\Pi(x, y)\{K \text{ indef.}\}$ , having a characteristic which is = the sum of the characteristics of the theta-functions. In fact, from the four

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equations of the theta-functions, we at once obtain for their products  $\Pi(x, y)$  the four equations

$$\begin{aligned}\Pi(x+1, y) &= (-)^{\Sigma\mu} \Pi(x, y), \\ \Pi(x, y+1) &= (-)^{\Sigma\nu} \Pi(x, y), \\ \Pi(x+A, y+H) &= (-)^{\Sigma q} \Pi(x, y) \exp. -i\pi K(2x+A), \\ \Pi(x+H, y+B) &= (-)^{\Sigma r} \Pi(x, y) \exp. -i\pi K(2y+B),\end{aligned}$$

which proves the theorem.

**14.** But if the indices are in direct relation to each other, then we have further

$$\Pi(-x, -y) = (-)^{\Sigma\mu q + \Sigma\nu r} \Pi(x, y),$$

and the product is thus a function  $\Pi(x, y)\{K, \text{def.}\}$ .

**15.** Take the square of a theta-function, the characteristic is  $= \begin{pmatrix} 0,0 \\ 0,0 \end{pmatrix}$  or 0, and we have also twice the index = 0; viz. the theta-function is in direct relation with itself. Hence the squared function is a function  $\Pi \begin{pmatrix} 0,0 \\ 0,0 \end{pmatrix} (x, y)\{2, \text{def.}\}$ , and as such it contains linearly  $\frac{1}{2}(2^2 + 4) = 4$  arbitrary constants. Hence, taking any five squares, since each of them is a function of the form in question, it follows that the squares of the 5 theta-functions are connected by a linear relation.

*Göpel's relation between 4 theta-functions.*

**16.** We may in a variety of ways (in fact, in 60 ways, as will presently be shown) select four theta-functions, all of them even, or else two of them even and two odd (that is, having the sum of their indices = 0), such that the sum of their characteristics is = 0; for instance, the functions may be

$$P = \begin{pmatrix} 0,0 \\ 0,0 \end{pmatrix}, \quad P' = \begin{pmatrix} 0,0 \\ 1,0 \end{pmatrix}, \quad S = \begin{pmatrix} 1,1 \\ 0,1 \end{pmatrix}, \quad S' = \begin{pmatrix} 1,1 \\ 1,1 \end{pmatrix},$$

indices

$$0, \quad 0, \quad 1, \quad 1.$$

The functions are thus in direct relation, and the product of the four functions is a function  $\Pi \begin{pmatrix} 0,0 \\ 0,0 \end{pmatrix} \{4, \text{def.}\}$ . But obviously any one of the functions taken four times, or any two of them taken each twice, are in like manner four functions in direct relation, or the fourth powers  $P^4, P'^4, S^4, S'^4$  and the squared products  $P^2 P'^2, P^2 S^2, P^2 S'^2, P'^2 S^2, P'^2 S'^2, S^2 S'^2$ , are in like manner each of them a function  $\Pi \begin{pmatrix} 0,0 \\ 0,0 \end{pmatrix} \{4, \text{def.}\}$ , viz. we have thus in all  $1 + 4 + \binom{4}{2} = 11$  such functions. But the  $\Pi$  function contains only  $\frac{1}{2}(4^2 + 4) = 10$  arbitrary constants; hence there must be a linear relation between the 11 powers and products, and this is Göpel's relation.

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**17.** Starting with any two characteristics  $a, b$  at pleasure, the remaining characteristics form seven pairs, such that

$$a + b = c + d = e + f = g + h = i + j = k + l = m + n = o + p;$$

but among the seven pairs we have only three, suppose  $c + d, e + f, g + h$ , which are such that  $(a, b, c, d), (a, b, e, f), (a, b, g, h)$  are each of them either all even or else two even and two odd; that is, starting with any pair  $(a, b)$ , we have these three tetrads having each of them the required property. The number of pairs  $(a, b)$  is  $16 \cdot 15 = 120$ ; and we thence derive  $120 \times 3 = 360$  tetrads; but each such tetrad is of course derivable from any one of the six pairs contained in it; or the number of distinct tetrads is  $360/6 = 60$ , viz. we have, as mentioned above, 60 Göpel-tetrads.

*The four functions  $\Pi_0, \Pi_1, \Pi_2, \Pi_3$ .*

**18.** We consider four theta-functions  $\theta_0, \theta_1, \theta_2, \theta_3$ , which are such that to the modulus 2, the sum of the characters is  $= 0$ , and also the sum of the indices is  $= 0$ ; taking the characters to be

$$\begin{pmatrix} \mu, \nu \\ q, r \end{pmatrix}, \begin{pmatrix} \mu', \nu' \\ q', r' \end{pmatrix}, \begin{pmatrix} \mu'', \nu'' \\ q'', r'' \end{pmatrix}, \begin{pmatrix} \mu''', \nu''' \\ q''', r''' \end{pmatrix},$$

and writing throughout  $=$  for  $\equiv (\text{mod } 2)$ , we have

$$\begin{aligned} \mu + \mu' + \mu'' + \mu''' &= 0, \\ \nu + \nu' + \nu'' + \nu''' &= 0, \\ q + q' + q'' + q''' &= 0, \\ r + r' + r'' + r''' &= 0, \\ \mu q + \nu r + \mu' q' + \nu' r' + \mu'' q'' + \nu'' r'' + \mu''' q''' + \nu''' r''' &= 0. \end{aligned}$$

Writing for shortness  $(01) = \mu q' + \mu' q + \nu r' + \nu' r$ , and so in other cases; and further  $(012) = (01) + (02) + (12)$ , then from  $(012)$ , &c., substituting  $\mu''', \nu''', q''', r'''$  their values for the first four equations, we deduce  $(012) = 0$ ; and similarly  $(013) = 0, (023) = 0, (123) = 0$ .

**19.** Consider now a product  $\theta_0^a \theta_1^b \theta_2^c \theta_3^d$ , where  $a + b + c + d$  is  $=$  a given odd number  $k$ ; the characteristic is

$$\begin{pmatrix} \mu a + \mu' b + \mu'' c + \mu''' d, & \nu a + \nu' b + \nu'' c + \nu''' d \\ q a + q' b + q'' c + q''' d, & r a + r' b + r'' c + r''' d \end{pmatrix},$$

and it hence follows that the index is

$$= a(\mu q + \nu r) + b(\mu' q' + \nu' r') + c(\mu'' q'' + \nu'' r'') + d(\mu''' q''' + \nu''' r''').$$

In fact, forming the index in question, we have first terms in  $a^2, b^2, c^2, d^2$ , which upon writing therein  $a, b, c, d$  for these values respectively ( $a^2 = a$ , &c.) give the

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required value; we have therefore only to show that the sum of the remaining terms in  $ab$ , &c., is  $= 0$ . These terms are

$$ab(01) + ac(02) + ad(03) + bc(12) + bd(13) + cd(23),$$

and writing herein  $a + b + c + d = 1$ , and thence  $ad = a(1 - a - b - c) = ab + ac$ , and similarly  $bd = ab + bc$ ,  $cd = ac + bc$ , the terms in question become

$$= ab(013) + ac(023) + bc(123),$$

that is, they become  $= 0$ .

**20.** We thus see that the function  $\theta_0^a \theta_1^b \theta_2^c \theta_3^d$  has an index which is

$$= a \text{ ind } \theta_0 + b \text{ ind } \theta_1 + c \text{ ind } \theta_2 + d \text{ ind } \theta_3.$$

Consider separately four products  $\theta_0^a \theta_1^b \theta_2^c \theta_3^d$ , in which the exponents  $a, b, c, d$  satisfy successively the relations (always to modulus 2)

$$\begin{aligned} b + d &= 0, & c + d &= 0, \\ b + d &= 1, & c + d &= 0, \\ b + d &= 0, & c + d &= 1, \\ b + d &= 1, & c + d &= 1. \end{aligned}$$

Combining herewith the relation  $a + b + c + d = 1$ , it follows that the exponents  $a, b, c, d$  are

$$\begin{array}{cccc} d+1, & d, & d, & d, \\ d, & d+1, & d, & d, \\ d, & d, & d+1, & d, \\ d, & d, & d, & d+1, \end{array}$$

in the four cases respectively. Then substituting these values, the characteristics become

$$\begin{pmatrix} \mu, \nu \\ q, r \end{pmatrix}, \begin{pmatrix} \mu', \nu' \\ q', r' \end{pmatrix}, \begin{pmatrix} \mu'', \nu'' \\ q'', r'' \end{pmatrix}, \begin{pmatrix} \mu''', \nu''' \\ q''', r''' \end{pmatrix},$$

viz. the four products have the same characters as  $\theta_0, \theta_1, \theta_2, \theta_3$  respectively; and in like manner recollecting that

$$\text{ind } \theta_0 + \text{ind } \theta_1 + \text{ind } \theta_2 + \text{ind } \theta_3 = 0,$$

we see that the four products have the same indices as  $\theta_0, \theta_1, \theta_2, \theta_3$  respectively.

More generally write  $\Pi = \Sigma A \theta_0^a \theta_1^b \theta_2^c \theta_3^d$ , where respectively the exponents  $a, b, c, d$  satisfy the conditions already referred to; then for the four cases  $\Pi_0, \Pi_1, \Pi_2, \Pi_3$  have the same characteristics, and the same indices, as  $\theta_0, \theta_1, \theta_2, \theta_3$  respectively.

**21.** It can be shown that each of the functions  $\Pi$  contains  $\frac{1}{2}(k^2 + 1)$  constants. It will be recollected that we have between  $\theta_0, \theta_1, \theta_2, \theta_3$  an equation of the form

$$0 = (\theta_0^4, \theta_1^4, \theta_2^4, \theta_3^4, \theta_0^2 \theta_1^2, \theta_0^2 \theta_2^2, \theta_0^2 \theta_3^2, \theta_1^2 \theta_2^2, \theta_1^2 \theta_3^2, \theta_2^2 \theta_3^2);$$

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this serves to express, say  $\theta_3^4$ , in lower powers of  $\theta_3$ ; and by successive applications of this equation, we can reduce  $\Sigma \theta_0^a \theta_1^b \theta_2^c \theta_3^d$  to a form in which  $d$  has only one of the values 0, 1, 2 or 3; we do not by this transformation alter the suffix of  $\Pi$ , viz. a term originally of the form  $\Pi_0, \Pi_1, \Pi_2$  or  $\Pi_3$ , will by the transformation give rise only to terms which are of the same form  $\Pi_0, \Pi_1, \Pi_2, \Pi_3$  (as the case may be). The number of constants in  $\Pi_0$  is thus

$$= \text{number of partitions of } k \text{ into four parts } a, b, c, d,$$

under the conditions

$$\begin{array}{ll} d = 0 \text{ or } 2; & a \text{ odd, } b, c \text{ each even,} \\ d = 1 \text{ or } 3; & a \text{ even, } b, c \text{ each odd,} \end{array}$$

where, in reckoning the partitions, the order of the parts is taken into account: the partitions are thus as follows

$$\begin{array}{ll} d = 0, & (a-1) + b + c = k-1, \\ d = 1, & a + (b-1) + (c-1) = k-3, \\ d = 2, & (a-1) + b + c = k-3, \\ d = 3, & a + (b-1) + (c-1) = k-5, \end{array}$$

where the parts  $a$  or  $(a-1)$ ,  $b$  or  $(b-1)$ ,  $c$  or  $(c-1)$ , as the case may be, are all of them even; hence, writing  $k' = \frac{1}{2}(k-1)$ , the cases are

$$a' + b' + c' = k', \quad k' - 1, \quad k' - 1, \quad \text{or } k' - 2,$$

where the  $a', b', c'$  are odd or even (zero not excluded) at pleasure; as already mentioned, the order of the parts is taken into account: thus the particulars of 3 would be the

$$300, 210, 120, 030, \quad \text{No. is } 10, = \frac{1}{2} \cdot 4 \cdot 5.$$

$$201, 111, 021,$$

$$102, 012,$$



003.

Hence, in the four cases respectively, the numbers are

$$\begin{aligned} & \frac{1}{2}(k' + 1)(k' + 2), \\ & \frac{1}{2}k'(k' + 1), \\ & \frac{1}{2}k'(k' + 1), \\ & \frac{1}{2}(k' - 1)k', \end{aligned}$$

giving a total

$$= \frac{1}{2}(4k'^2 + 4k' + 2) = \frac{1}{2}\{(2k' + 1)^2 + 1\},$$

that is,  $= \frac{1}{2}(k^2 + 1)$ . And similarly the number is  $= \frac{1}{2}(k^2 + 1)$  in the other three cases respectively.

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*The Hermitian quartic matrix.*

**22.** Observing that, for the adopted form of the  $\Pi$ - or  $\Theta$ -functions, the periods  $\omega_0, v_0$  are 1, 0,

$$\begin{aligned} \omega_1, v_1 &= 0, 1, \\ \omega_2, v_2 &= A, H, \\ \omega_3, v_3 &= H, B, \end{aligned}$$

so that we have

$$\omega_0 v_1 - \omega_1 v_0 + \omega_2 v_3 - \omega_3 v_2 = H - 0 + 0 - H = 0,$$

we have to consider the automorphic transformation of the bilinear form

$$\omega_0 v_1 - \omega_1 v_0 + \omega_2 v_3 - \omega_3 v_2.$$

**23.** We write

$$\begin{aligned} (\omega_0, \omega_1, \omega_2, \omega_3) &= (a_0, a_1, a_2, a_3 \mid \Omega_0, \Omega_1, \Omega_2, \Omega_3) \mid b_0, b_1, b_2, b_3 \mid c_0, c_1, c_2, c_3 \mid d_0, d_1, d_2, d_3, \\ (v_0, v_1, v_2, v_3) &= (c_0, c_1, c_2, c_3 \mid T_0, T_1, T_2, T_3) \mid d_0, d_1, d_2, d_3 \mid a_0, a_1, a_2, a_3 \mid b_0, b_1, b_2, b_3, \end{aligned}$$

and the coefficients are assumed to be such that we have identically

$$\omega_0 v_1 - \omega_1 v_0 + \omega_2 v_3 - \omega_3 v_2 = k(\Omega_0 T_1 - \Omega_1 T_0 + \Omega_2 T_3 - \Omega_3 T_2),$$

where  $k$  is in the sequel taken to be a positive integer. We obtain by direct substitution the value of  $\omega_0 v_1 - \omega_1 v_0 + \omega_2 v_3 - \omega_3 v_2$  in the following form (with a  $4 \times 4$  array of bilinear coefficients  $\Omega_i T_j$ ):

[a  $4 \times 4$  matrix of expressions of the form  $a_i c_j - a_j c_i + b_i d_j - b_j d_i$ , indexed by the  $\Omega_i T_j$  pairs.]

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viz. equating this to its value  $k(\Omega_0 T_1 - \Omega_1 T_0 + \Omega_2 T_3 - \Omega_3 T_2)$ , we have 4 identities and 12 equations which are, in fact, 6 equations occurring each twice. We have thus six equations, which are the conditions in order that the matrix may be the matrix of automorphic transformation of the bilinear form  $\omega_0 v_1 - \omega_1 v_0 + \omega_2 v_3 - \omega_3 v_2$ . The six equations may be written

$$\begin{aligned} (ac + bd)_{01} &= 0, \\ (ac + bd)_{02} &= k, \\ (ac + bd)_{03} &= 0, \\ (ac + bd)_{12} &= 0, \\ (ac + bd)_{13} &= k, \\ (ac + bd)_{23} &= 0, \end{aligned}$$

viz. the first of these equations is  $a_0c_1 - a_1c_0 + b_0d_1 - b_1d_0 = 0$ , and so in other cases. It is convenient to remark that each interchange of two letters  $a$  and  $c$ ,  $b$  and  $d$ , also interchange of the suffixes 0 and 2, produces a change of sign; thus the second equation may be written  $(ca + db)_{02} = k$ , or  $(ac + bd)_{20} = -k$ .

**24.** The inverse matrix is found to be

$$\begin{pmatrix} a_0, a_1, a_2, a_3 \\ b_0, b_1, b_2, b_3 \\ c_0, c_1, c_2, c_3 \\ d_0, d_1, d_2, d_3 \end{pmatrix}^{-1} = \frac{1}{k} \begin{pmatrix} c_2, d_2, -a_2, -b_2 \\ c_3, d_3, -a_3, -b_3 \\ -c_0, -d_0, a_0, b_0 \\ -c_1, -d_1, a_1, b_1 \end{pmatrix},$$

and the determinant of each of the matrices in this formula is  $= k^2$ .

We have thus

$$(\Omega_0, \Omega_1, \Omega_2, \Omega_3) = \frac{1}{k}(c_2, d_2, -a_2, -b_2 \mid \omega_0, \omega_1, \omega_2, \omega_3), \quad \text{etc.,}$$

and the like formula for the  $T, v$ . Substituting these values in the equation

$$k(\Omega_0 T_1 - \Omega_1 T_0 + \Omega_2 T_3 - \Omega_3 T_2) = \omega_0 v_1 - \omega_1 v_0 + \omega_2 v_3 - \omega_3 v_2,$$

we obtain 6 new equations, which are in a different form the conditions for the automorphic transformation.

The 6 new equations may be written

$$\begin{aligned} (02 + 13)_{ab} &= 0, \\ (02 + 13)_{ac} &= k, \\ (02 + 13)_{ad} &= 0, \\ (02 + 13)_{bc} &= 0, \\ (02 + 13)_{bd} &= k, \\ (02 + 13)_{cd} &= 0, \end{aligned}$$

viz. these equations are  $a_0c_2 - c_0a_2 + a_1c_3 - c_1a_3 = 0$ , &c.

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**25.** It is worth while to show how the foregoing formula for the inverse matrix comes out. Take for instance the diagonal minor

$$\begin{vmatrix} b_1, b_2, b_3 \\ c_1, c_2, c_3 \\ d_1, d_2, d_3 \end{vmatrix},$$

this is

$$c_1(bd)_{23} + c_2(bd)_{31} + c_3(bd)_{12},$$

which is

$$= c_1(ac)_{23} + c_2\{k + (ac)_{31}\} + c_3(ac)_{12},$$

since the remaining terms destroy each other. And dividing by the determinant, which is  $= k^2$ , we have the term  $c_2 \div k$  of the inverse matrix.

*The Symmetrical Hermitian Matrix.*

**26.** We may consider a symmetrical Hermitian matrix, say the matrix

$$\begin{pmatrix} \mathfrak{A}, \mathfrak{H}, \mathfrak{G}, \mathfrak{I} \\ \mathfrak{H}, \mathfrak{B}, \mathfrak{F}, \mathfrak{M} \\ \mathfrak{G}, \mathfrak{F}, \mathfrak{C}, \mathfrak{N} \\ \mathfrak{I}, \mathfrak{M}, \mathfrak{N}, \mathfrak{D} \end{pmatrix},$$

viz. we have

$$\begin{aligned}\mathfrak{A}\mathfrak{G} - \mathfrak{H}^2 + \mathfrak{H}\mathfrak{N} - \mathfrak{T}\mathfrak{F} &= \phi, \\ \mathfrak{H}\mathfrak{N} - \mathfrak{T}\mathfrak{B} + \mathfrak{T}\mathfrak{D} - \mathfrak{M}^2 &= \phi, \\ \mathfrak{A}\mathfrak{B} - \mathfrak{H}\mathfrak{G} + \mathfrak{H}\mathfrak{N} - \mathfrak{T}\mathfrak{F} &= 0, \\ \mathfrak{A}\mathfrak{M} - \mathfrak{T}\mathfrak{H} + \mathfrak{H}\mathfrak{D} - \mathfrak{T}\mathfrak{N} &= 0, \\ \mathfrak{G}\mathfrak{D} - \mathfrak{N}\mathfrak{G} + \mathfrak{B}\mathfrak{N} - \mathfrak{M}\mathfrak{G} &= 0, \\ \mathfrak{B}\mathfrak{N} - \mathfrak{G}\mathfrak{F} + \mathfrak{G}\mathfrak{M} - \mathfrak{N}\mathfrak{F} &= 0.\end{aligned}$$

The characteristic property is that, effecting a Hermitian transformation, we have a new symmetrical Hermitian matrix

$$\begin{pmatrix} \mathfrak{A}', \mathfrak{H}', \mathfrak{G}', \mathfrak{T}' \\ \mathfrak{H}', \mathfrak{B}', \mathfrak{F}', \mathfrak{M}' \\ \mathfrak{G}', \mathfrak{F}', \mathfrak{C}', \mathfrak{N}' \\ \mathfrak{T}', \mathfrak{M}', \mathfrak{N}', \mathfrak{D}' \end{pmatrix} = (\mathfrak{A}', \mathfrak{H}', \mathfrak{G}', \mathfrak{T}' \mid x, y, z, w)^2.$$

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In fact, the new matrix is

$$(\mathfrak{A}' \dots) = \begin{pmatrix} a_0, b_0, c_0, d_0 \\ a_1, b_1, c_1, d_1 \\ a_2, b_2, c_2, d_2 \\ a_3, b_3, c_3, d_3 \end{pmatrix} (\mathfrak{A} \dots) \begin{pmatrix} a_0, a_1, a_2, a_3 \\ b_0, b_1, b_2, b_3 \\ c_0, c_1, c_2, c_3 \\ d_0, d_1, d_2, d_3 \end{pmatrix},$$

where the first factor, qua transposed matrix, is Hermitian. Hence the product is a symmetrical  $\frac{k}{\square}\Omega$  matrix.

27. Write

$$\begin{aligned}\Delta_0 &= H_0^2 - A_0B_0 - (\eta^2 - a\beta), \quad \theta = a\beta - \eta^2, \\ \delta &= 2H_0\eta - aB_0 - \beta A_0,\end{aligned}$$

and consider the following matrix

$$\begin{aligned}(\beta, -\eta, -\eta H_0 + \beta A_0, -\eta B_0 + \beta H_0 \mid x, y, z, w)^2 \\ -\eta, a, -\eta A_0 + aH_0, -\eta H_0 + aB_0, \\ -\eta H_0 + \beta A_0, -\eta A_0 + aH_0, a\Delta_0 - \delta A_0, -\delta H_0 + \eta\Delta_0, \\ -\eta B_0 + \beta H_0, -\eta H_0 + aB_0, \delta H_0 - \eta\Delta_0, \beta\Delta_0 - \delta B_0,\end{aligned}$$

which is

$$= (\beta, -\eta, a\eta x + A_0y + \beta z + H_0w)^2 + \frac{1}{\theta}(\eta a - \beta\eta)(az - \beta z + \dots);$$

it is easily shown that this is a symmetrical  $\theta$ -matrix. In fact, representing it for a moment by

$$\begin{aligned}(\mathfrak{A}, \mathfrak{H}, \mathfrak{G}, \mathfrak{T}) &\dots \\ (\mathfrak{H}, \mathfrak{B}, \mathfrak{F}, \mathfrak{M}) &\dots \\ (\mathfrak{G}, \mathfrak{F}, \mathfrak{C}, \mathfrak{N}) &\dots \\ (\mathfrak{T}, \mathfrak{M}, \mathfrak{N}, \mathfrak{D}) &\dots\end{aligned}$$

so that  $\mathfrak{A} = \beta$ ,  $\mathfrak{B} = a$ ,  $\mathfrak{H} = -\eta$ , &c., we have

$$\mathfrak{A}\mathfrak{G} - \mathfrak{H}^2 + \mathfrak{H}\mathfrak{N} - \mathfrak{T}\mathfrak{F} = \beta(a\Delta_0 - \delta A_0) - (-\eta H_0 + \beta A_0)^2 - \eta(-\delta H_0 + \eta\Delta_0) - (-\eta B_0 + \beta H_0)(-\eta A_0 + aH_0),$$

which is

$$= a\beta\Delta_0 - \beta\delta A_0 - \eta^2 H_0^2 + 2\beta\eta A_0 H_0 - \beta^2 A_0^2 + \eta\delta H_0 - \eta^2\Delta_0 + \eta^2 B_0 A_0 - \eta a B_0 H_0 + \eta\beta A_0 H_0 - a\beta H_0^2;$$

or, for the two terms  $\beta\delta A_0 + \eta\delta H_0 = (\eta H_0 - \beta A_0)\delta$ , substituting the value  $\delta = 2H_0\eta - aB_0 - \beta A_0$ , the whole is found to be

$$= (a\beta - \eta^2)(\Delta_0 - H_0^2 + A_0B_0), \quad \text{that is,} \quad = \theta(-\theta) = -\theta^2.$$

And, similarly,  $\mathfrak{H}\mathfrak{N} - \mathfrak{T}\mathfrak{B} + \mathfrak{T}\mathfrak{D} - \mathfrak{M}^2$  is found to be  $= -\theta^2$ ; and the remaining four combinations of terms to be each of them  $= 0$ ; the matrix is thus a symmetrical  $\theta$ -matrix.

**28.** It is to be added that the diagonal minors and the determinant

$$\mathfrak{A}, \mathfrak{A}\mathfrak{B} - \mathfrak{H}^2, \mathfrak{A}\mathfrak{B}\mathfrak{C} + \&c., \mathfrak{A}\mathfrak{B}\mathfrak{C}\mathfrak{D} + \&c.,$$

have respectively the values  $\beta, \theta, a\theta^2, \theta^4$ , viz. if  $a, \beta, \theta$  are positive, then these are all positive; or the last-mentioned quadric function is a definite positive form.

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*The general matrix resumed: Arithmetical theory.*

**29.** The matrix containing, as before, the parameter  $k$ , may be called a  $k$ -matrix; if  $k$  be  $= 1$ , it is a unit-matrix. From the fundamental equation

$$k(\Omega_0 T_1 - \Omega_1 T_0 + \Omega_2 T_3 - \Omega_3 T_2) = \omega_0 v_1 - \omega_1 v_0 + \omega_2 v_3 - \omega_3 v_2,$$

it at once follows that, compounding a  $k$ -matrix with a  $k'$ -matrix, we have a  $kk'$ -matrix; and in particular, compounding a  $k$ -matrix with a unit-matrix, we have a  $k$ -matrix.

**30.** The symbols  $a_0, a_1, b_0, \&c.$ , and  $k$ , have thus far been arbitrary magnitudes, but we now take them to be integers; and we consider in particular the case where  $k$  is a positive odd prime. The number of  $k$ -matrices is of course infinite, but if we regard as equivalent any two such matrices which are derivable one from the other by post-multiplication by a unit-matrix (viz.  $U$  being a unit-matrix, the matrices  $M$  and  $M \cdot U$  are regarded as equivalent), then the number of distinct  $k$ -matrices is finite, and  $= 1 + k + k^2 + k^3$ .

The first step is to show that we can by post-multiplication, by a properly determined unit-matrix, reduce the  $k$ -matrix to the form

$$\begin{pmatrix} a_0, a_1, a_2, a_3 \\ 0, b_1, b_2, b_3 \\ 0, 0, c_2, c_3 \\ 0, 0, d_2, d_3 \end{pmatrix},$$

these values being such as to satisfy identically two out of the six conditions; the remaining conditions present themselves under the two equivalent forms

$$a_0 c_2 = k, \quad b_1 d_3 = k, \quad a_0 b_2 - a_2 b_0 = 0, \quad a_0 d_1 - a_1 d_0 = 0,$$

and

$$a_0 c_2 = k, \quad b_1 d_3 = k, \quad a_0 c_1 - a_1 c_0 = 0, \quad b_2 d_3 - b_3 d_2 = 0.$$

Hence  $a_0, c_2 = 1, k$ , or  $k, 1$ ; and  $b_1, d_3 = 1, k$ , or  $k, 1$ ; so that, combining these pairs of values, we have four different types of matrix, each type depending on the coefficients  $a_1, a_2, a_3, b_2, b_3, c_3$ , connected together by two equations. But the forms of the same type are not distinct from each other, and we have to determine for each of the four types a system of non-equivalent forms comprised therein, and such that from these, by post-multiplication by a unit-matrix as before, the other forms of the type can be obtained. This final system is:

$$\begin{array}{ll} \text{I.} & \begin{pmatrix} 1, 0, 0, 0 \\ 0, 1, 0, 0 \\ 0, 0, k, 0 \\ 0, 0, 0, k \end{pmatrix}, & \text{II.} & \begin{pmatrix} 1, 0, 0, 0 \\ 0, k, 0, i \\ 0, 0, k, 0 \\ 0, 0, 0, 1 \end{pmatrix}, \\ \text{III.} & \begin{pmatrix} k, i, i', 0 \\ 0, 1, 0, 0 \\ 0, 0, 1, 0 \\ 0, 0, -i, k \end{pmatrix}, & \text{IV.} & \begin{pmatrix} k, 0, i, i' \\ 0, k, i, i'' \\ 0, 0, 1, 0 \\ 0, 0, 0, 1 \end{pmatrix}, \end{array}$$

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where  $i, i', i''$  are integers, having each of them any one of the values  $0, 1, 2, \dots, k-1$ , viz. there are in the four types  $k-1, k, k^2, k^3$  forms respectively, and the number of forms is thus  $= 1 + k + k^2 + k^3$ , as already mentioned. I abstain from the further details of the proof.

There is obviously a like theory for which, in place of post-multiplication, we have pre-multiplication; viz. here the matrices  $M$  and  $U \cdot M$  are regarded as equivalent.

**31.** Any two  $k$ -matrices are reducible one to the other by a combined pre- and post-multiplication; viz. we have always  $M' = U \cdot M \cdot U'$ , where  $M, M'$  are any two  $k$ -matrices, and  $U, U'$  being properly determined unit-matrices; and in particular,  $M$  any given  $k$ -matrix, this is expressible in the foregoing form, where  $M$  denotes the principal matrix

$$\begin{pmatrix} 1, & 0, & 0, & 0 \\ 0, & 1, & 0, & 0 \\ 0, & 0, & k, & 0 \\ 0, & 0, & 0, & k \end{pmatrix}.$$

*Congruence theorems,  $k$  an odd number.*

**32.** Taking  $k$  an odd number, and using throughout  $=$  instead of  $\equiv (\text{mod } 2)$ , we have the following congruences:

$$(a_0c_0 + b_0d_0, a_0c_1 + b_0d_1, a_0c_2 + b_0d_2, a_0c_3 + b_0d_3) = (a_0, a_1, a_2, a_3 \mid \Sigma a_0c_0 + b_0d_0, a_0c_1 + b_0d_1, a_0c_2 + b_0d_2, a_0c_3 + b_0d_3 \mid b_0, b_1, b_2, b_3 \mid c_0, c_1, c_2, c_3 \mid d_0, d_1, d_2, d_3),$$

or, conversely,

$$(a_0c_0 + b_0d_0, a_0c_1 + b_0d_1, a_0c_2 + b_0d_2, a_0c_3 + b_0d_3) = (c_0, -d_0, -a_0, -b_0 \mid \Sigma \dots \mid c_1 \dots \mid c_2 \dots \mid c_3 \dots),$$

where observe that on the right-hand side the signs  $-$  may be changed into  $+$ ; in fact, to the modulus 2, we have for any integer value whatever  $-p = +p$ .

**33.** The first congruence is

$$\begin{aligned} a_2(a_0c_0 + b_0d_0), \text{ say } &= Y, \\ &= a_0(a_0c_2 + \dots), \text{ say } = X, \end{aligned}$$

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and to verify this, I see no other method than that of considering separately all the combinations of even and odd values of  $a_0, a_1, a_2, a_3$ ; viz. we have the 16 cases

| Case | $a_0$ | $a_1$ | $a_2$ | $a_3$ | $X$            | $Y$ |
|------|-------|-------|-------|-------|----------------|-----|
| 1    | 0     | 0     | 0     | 0     | does not exist |     |
| 2    | 0     | 0     | 0     | 1     | 0              | 0   |
| 3    | 0     | 0     | 1     | 0     | 0              | 0   |
| 4    | 0     | 0     | 1     | 1     | 1              | 1   |
| 5    | 0     | 1     | 0     | 0     | 0              | 0   |
| 6    | 0     | 1     | 0     | 1     | 0              | 0   |
| 7    | 0     | 1     | 1     | 0     | 0              | 0   |
| 8    | 0     | 1     | 1     | 1     | 1              | 1   |
| 9    | 1     | 0     | 0     | 0     | 0              | 0   |
| 10   | 1     | 0     | 0     | 1     | 0              | 0   |
| 11   | 1     | 0     | 1     | 0     | 1              | 1   |
| 12   | 1     | 0     | 1     | 1     | 1              | 1   |
| 13   | 1     | 1     | 0     | 0     | 1              | 1   |
| 14   | 1     | 1     | 0     | 1     | 1              | 1   |
| 15   | 1     | 1     | 1     | 0     | 1              | 1   |
| 16   | 1     | 1     | 1     | 1     | 0              | 0   |

viz. the  $X$  column gives in each case the value of  $X$ ,  $= a_0c_2 + a_1c_3$ , and then it has to be shown that  $Y$  has the same value. The coefficients satisfy the conditions

$$\begin{aligned} a_0c_1 - a_1c_0 + b_0d_1 - b_1d_0 &= 1, \\ b_0d_1 - b_1d_0 + b_2d_3 - b_3d_2 &= 0, \\ a_0b_1 + a_1b_0 + a_2b_3 - a_3b_2 &= 0, \\ a_0d_1 - a_1d_0 + a_2d_3 - a_3d_2 &= 0, \\ b_0c_1 - b_1c_0 + b_2c_3 - b_3c_2 &= 0, \\ c_0d_1 - c_1d_0 + c_2d_3 - c_3d_2 &= 0, \end{aligned}$$

so that from the first equation we cannot have  $a_0, a_1, a_2, a_3$  each  $= 0$ , or the first case does not exist. As to the remaining cases, it is easy to see that they group themselves as follows: 2, 3, 5, 9; 4, 13; 6, 7, 10, 11; 8, 12, 14, 15; 16: the proof being substantially the same for the several cases in the same group.

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**34.** Case 2. We have  $Y = b_0d_1$ , which should be  $= 0$ , and in fact, the six equations give  $b_0 = 0, d_1 = 0$ , whence  $Y = 0$ .

Case 4. We have  $Y = c_0 + c_1 + b_0d_1 + b_0d_1$ , which should  $= 1$ , and in fact, the six equations give  $c_3 - c_2 = 1, b_3 - b_2 = 0, d_3 - d_2 = 0$ ; whence  $b_2d_3 - b_3d_2 = 1, d_3b_2 - b_3d_2 = 0$ , and therefore  $Y = 1$ .

Case 6. Here  $Y = b_0d_1 + d_0b_1$ , which should  $= 0$ ; and in fact, the six equations give  $b_0 + b_1 = 0, d_0 + d_1 = 0$ , whence  $Y = 0$ .

Case 16. Here  $Y = c_0 + d_0 + c_2 + c_3 + \mu_0 + \mu_1 + \mu_2 + \mu_3$ , which should be  $= 0$ ; the six equations give

$$\begin{aligned} -c_0 - d_0 + c_2 + c_3 &= 1, & -b_0 - b_1 + b_2 - b_3 &= 0, \\ -d_0 - d_1 + d_2 + d_3 &= 0, & b_2d_3 + b_3d_0 + b_1d_3 - b_3d_1 &= 1. \end{aligned}$$

Writing  $\mu = b_0 + b_1 + b_2 + b_3$  and  $\mu' = d_0 + d_1 + d_2 + d_3$ , we find

$$\mu\mu' = b_0d_0 + b_1d_1 + b_2d_2 + b_3d_3 + (b_0 + b_2)(d_1 + d_3) + (b_1 + b_3)(d_0 + d_2),$$

that is,

$$\mu\mu' = b_0d_0 + b_1d_1 + b_2d_2 + b_3d_3 + 1,$$

and  $c + c_1 + c_2 + c_3 = 1$ ; whence  $Y = 0$ .

**35.** Case 8. This is the only case of any difficulty: we have

$$Y = c_0 + c_2 + b_0d_1 + b_0d_1 + b_0d_1,$$

which should be  $= 1$ . The six equations give

$$-c_0 + c_2 - c_1 = 1, \quad -b_0 + b_2 - b_1 = 0, \quad -d_0 + d_1 - d_0 = 0,$$

or, say

$$c_0 = -1 - c_1 + c_2, \quad b_0 = -b_1 + b_2, \quad d_0 = d_1;$$

or, substituting these values and omitting even terms

$$Y = d_1 + c_2 - d_0b_2 - b_2d_1.$$

The remaining three of the six equations are

$$\begin{aligned} -b_2d_0 + b_0d_2 - d_0b_1 + b_1d_0 + b_2d_3 - b_3d_2 &= 1, \\ a_0b_2 - b_0 + c_0d_1 - c_2d_0 &= 0, \\ c_1d_3 - c_3d_1 &= 0; \end{aligned}$$

or, substituting for  $b, c, d$  their values, these become

$$(1 + c_1 - c_2)b_2 - (b_3 - b_1)c_2 = -\mu_1 - d_2 + b_2c_1,$$

$$\begin{aligned} -(d_1 - d_3)b_2 - \mu_3 &= -(b_3 - b_1)d_3 - 1 + \mu_1, \\ -c_1 - c_3 + c_2(c_3 + c_2) - d_2(-1 + c_2)d_2 &= -d_3d_1 + d_2d_3; \end{aligned}$$

we can from these equations eliminate  $b_2, c_2, d_2$ ; viz. from the first and third equations eliminating  $c_2$ , we have

$$\{(d_1 - d_3)b_2 + \mu_3\}(1 - c_0) - (d_3 - d_1)(b_1 + b_3 - b_0d_0) = -(\mu_1 + \mu_2)b_2 + (b_3 - b_1)(-c_3 + c_2d_1 - c_1d_1),$$

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and this, by means of the second equation, becomes

$$\mu_3 - \mu_0 + (1 - c_1)d_1 = (d_3 - d_1)(-b_1c_3 + b_3c_1) + (b_3 - b_1)(-c_3d_1 + c_1d_3),$$

viz. reducing, this is

$$\mu_3 - \mu_0 + d_1 = b_3c_1 - 1 + c_1;$$

and in virtue of it we have  $Y = c_2 + c_3 + b_1c_1 + 1$ .

It can be further shown that, to the modulus 2 ( $k$  being, as before, odd), we have

$$\begin{aligned} (a_0c_2 + a_1c_3)(c_0c_2 + c_1c_3) + (b_0d_2 + b_1d_3)(d_0d_2 + d_1d_3) &= 0, \\ (a_0c_0 + b_0d_0) + (a_1c_1 + b_1d_1) + (a_2c_2 + b_2d_2) + (a_3c_3 + b_3d_3) &= 0. \end{aligned}$$

To prove the first equation, write for a moment

$$X = (a_0c_2 - a_2c_0)(a_1c_3 - a_3c_1), \quad X' = (b_0d_2 - b_2d_0)(b_1d_3 - b_3d_1);$$

then in virtue of the equations we have  $X = X'$ . But we have identically

$$\Omega X = -X + (a_0c_2 + a_1c_3)(a_2c_0 + a_3c_1) \cdot \Omega$$

and from the equation

$$a_0c_2 + a_2c_0 + a_1c_3 + a_3c_1 = 1,$$

that is,  $a_2c_0 + a_3c_1 = 1 - a_0c_2 - a_1c_3$ , we have  $\Omega X + X = 0$ ; and similarly  $\Omega' X' + X' = 0$ , that is,  $\Omega X - \Omega' X' = 0$ ; we have thus the required equation  $\Omega X + \Omega' X' = 0$ . In a similar manner the second equation may be verified.

**36.** Write

$$\begin{aligned} p &= \mu a_0 + \nu a_1 + q a_2 + r a_3 + a_0 a_2 + a_1 a_3, \\ p' &= \mu b_0 + \nu b_1 + q b_2 + r b_3 + b_0 b_2 + b_1 b_3, \\ q' &= \mu c_0 + \nu c_1 + q c_2 + r c_3 + c_0 c_2 + c_1 c_3, \\ r' &= \mu d_0 + \nu d_1 + q d_2 + r d_3 + d_0 d_2 + d_1 d_3. \end{aligned}$$

It is to be shown that to the modulus 2 we have

$$\mu' q' + \nu' r' = \mu q + \nu r.$$

In fact, forming the value of  $\mu' q' + \nu' r'$ , we have first a constant term (term without  $\mu, \nu, q, r$ ) which vanishes; next writing  $\mu^2 = \mu$ , the whole term in  $\mu$  is

$$a(d_0d_2 + c_1c_3) + b(d_0d_2 + a_1a_3) + c(a_0a_2 + d_1d_3) + d(b_0b_2 + b_1b_3),$$

which also vanishes; and similarly the terms in  $\nu, q, r$  each of them vanish; there remain only the terms in  $\mu\nu, \mu q$ , &c. The coefficient of  $\mu q$  is  $a_0c_2 + a_2c_0 + b_0d_2 + b_2d_0$ ,

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which (always to the modulus 2) is  $= a_0c_2 + a_2c_0 + b_0d_2 + b_2d_0$ , that is, it is  $= 1$ ; and similarly the coefficient of  $\nu r$  is  $= 1$ ; and in like manner the coefficients of the other terms are each  $= 0$ ; we have thus the required congruence  $\mu' q' + \nu' r' = \mu q + \nu r$ .

*The quintic matrix.*

**37.** I consider a quintic matrix composed of the foregoing coefficients  $(a, b, c, d)_{0,1,2,3}$ , viz. the matrix contained in a linear transformation which I write as follows:

$$(T', P', Q', R', S') = \begin{pmatrix} \cdot & 01 & 2\cdot 02 & 03 & 32 \\ ab & & & & \\ \cdot & & & & \\ cb & & & & \\ ac & & * & & \\ ad & & & * & \\ dc & & & & \end{pmatrix} (T, P, Q, R, S),$$

read

$$T' = T + (ab)_{01}P + 2(ab)_{02}Q + (ab)_{03}R + (ab)_{32}S, \\ P' = \dots \quad (\text{and so on}),$$

where as before  $(ab)_{01} = a_0b_1 - a_1b_0$ , &c., and so in other cases; in particular, observe that, in the expression of  $Q'$ , the term involving  $Q$  is  $\{k + 2(ac)_{02}\}Q$  which, in virtue of the relation  $(ac + bd)_{02} = k$ , may also be written  $\{(ac)_{02} - (bd)_{02}\}Q$ . I notice that in Hermite's paper, p. 366, the term is in effect written without the  $= 2k$ ,  $Q(ac)_{02}$ : the correction of this erratum and of a corresponding one, p. 366, is made p. 787 at the conclusion of the memoir.

**38.** The matrix is automorphic for the form  $T^2 - PR - TS$ , viz. we have identically

$$Q'^2 - P'R' - T'S' = k(Q^2 - PR - TS).$$

As a partial verification, consider in  $Q'^2 - P'R' - T'S'$  the term containing  $Q^2$ . The coefficient of  $Q^2$  is

$$\{-k + 2(ac)_{02}\}^2 - 4(ab)_{02}(ad)_{12} - 4(ab)_{02}(dc)_{02},$$

where the first term is

$$k^2 - 4k(ac)_{02} + 4(ac)_{02}^2;$$

Hence, observing that we have  $(ac + bd)_{02} = k$ , the whole coefficient is  $k^2$ , as it should be; and in like manner the verification may be effected for any other term.

**39.** We require the following formulae:

$$(-c_0, a_0, b_0)(T', P', Q') = k(\dots, b_1, -b_2, -b_3 \mid T, P, Q, R, S), \\ -c_1, a_1, b_1 : \quad \dots, b_1, b_2, b_3, \\ -c_2, a_2, b_2 : \quad b_1, b_2, b_3, \\ -c_3, a_3, b_3 : \quad b_3, b_2, b_1,$$

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that is,

$$-c_0T' + a_0P' + b_0Q' = k(b_1P - Qb_2 - b_3S), \quad \&c.,$$

and

$$(-d_0, a_0, b_0)(T', Q', R') = k(-a_0, -a_1, a_0, a_2, a_3 \mid T, P, Q, R, S), \\ -d_2, a_2, b_2 : \quad a_0, a_1, a_2, a_3, \\ -d_3, a_3, b_3 : \quad -a_0, a_1, a_2, a_3.$$

From the first set, multiplying the first, third and fourth equations by  $T, P, Q$  respectively and adding, and again multiplying the second, third and fourth equations by  $T, Q, R$ , and adding, we obtain

$$-(c_1T + c_2P + c_3Q)T + (a_1T + a_2P + a_3Q)P + (b_1T + b_2P + b_3Q)Q = kb_3(Q^2 - PR - TS),$$



$$(\text{similarly for } R) = -kb_2(Q^2 - PR - TS);$$

and similarly, from the second set of equations, we obtain

$$\begin{aligned} -(d_1T + d_2P + d_3Q)T + (a_0T + a_2P + a_3Q)Q + (b_0T + b_2P + b_3Q)R &= -ka_3(Q^2 - PR - TS), \\ -(d_1T + d_2Q + d_3R)T + (a_1T + a_2Q + a_3R)Q + (b_1T + b_2Q + b_3R)R &= ka_2(Q^2 - PR - TS). \end{aligned}$$

40. We have the inverse system

$$(T, P, Q, R, S) = \begin{pmatrix} \cdot & dc & ad & 2bd & cb & ab \\ 03 & & & & & \\ 13 & \cdot & -k & & & \\ 21 & & & & & \\ 01 & & & & & \end{pmatrix} (T', P', Q', R', S'),$$

read

$$T = T' + (dc)_{03}P' + (ad)_{31}Q' + (bd)_{31}R + (cb)_{21} + (ab)^*S';$$

and in particular observe that, in the expression for  $Q$ , the term containing  $Q'$  is  $Q'\{-k + 2(bd)_{13}\}$ , where, in virtue of  $(ac + bd)_{13} = k$ , the coefficient of  $Q'$  is also  $= (bd)_{13} - (ac)_{13}$ .

41. These equations give

$$(a_0, a_2, a_3 \mid T', P', Q') = k(\dots, d_3, -a_3, -b_3, \dots \mid T, P, Q, R, S),$$

$$b_0, b_2, b_3 : \quad -c_3, a_3, b_3,$$

$$c_0, c_2, c_3 : \quad \dots, d_3, c_3, \dots, b_3,$$

$$d_0, d_2, d_3 : \quad \dots, \dots, d_3, c_3, a_3,$$

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and

$$(a_1, a_2, a_3 \mid T', Q', R') = k(-d_2, \dots, a_2, b_2, \dots \mid T, P, Q, R, S),$$

$$b_1, b_2, b_3 : \quad c_2, -a_2, -b_2, \dots,$$

$$c_1, c_2, c_3 : \quad \dots, -d_2, c_2, \dots, -b_2,$$

$$d_1, d_2, d_3 : \quad \dots, \dots, d_2, c_2, a_2.$$

From the first set of equations, multiplying the first, third and fourth equations by  $-S'$ ,  $-R'$ ,  $+Q'$  respectively, and adding, and again multiplying the second, third and fourth equations by  $-S'$ ,  $Q'$ ,  $-P'$  respectively, and adding, we deduce

$$T(-a_0S' - c_0R' + d_0Q') + P(-a_2S' - c_2R' + d_2Q') + Q(-a_3S' - c_3R' + d_3Q') = -kc_3(Q'^2 - P'R' - T'S'),$$

$$T(-b_0S' + c_0Q' - d_0P') + P(-b_2S' + c_2Q' - d_2P') + Q(-b_3S' + c_3Q' - d_3P') = kd_3(Q'^2 - P'R' - T'S');$$

and in like manner, from the second set of equations,

$$T(-a_1S' - c_1R' + d_1Q') + Q(-a_2S' - c_2R' + d_2Q') + R(-a_3S' - c_3R' + d_3Q') = -kd_2(Q'^2 - P'R' - T'S'),$$

$$T(-b_1S' + c_1Q' - d_1P') + Q(-b_2S' + c_2Q' - d_2P') + R(-b_3S' + c_3Q' - d_3P') = kc_2(Q'^2 - P'R' - T'S').$$

42. Assume that  $T, P, Q, R, S$  and  $T', P', Q', R', S'$  are linearly connected as above; and write

$$T : P : Q : R : S = 1 : A : H : B : H^2 - AB,$$

$$T' : P' : Q' : R' : S' = 1 : A' : H' : B' : H'^2 - A'B',$$

equations which establish a like relation between  $1, A, H, B, H^2 - AB$ , and  $1, A', H', B', H'^2 - A'B'$ . Observe that these forms are admissible since, if

$$T^2 - PR - QS = 0,$$

then also

$$Q'^2 - P'R' - T'S' = 0.$$

The quantities  $A, H, B$  were taken as the parameters of a theta-function; viz. taking

$$A, H, B = A_0 + ia, H_0 + i\eta, B_0 + i\beta,$$

then  $(a, \eta, \beta \mid x, y)^2$  must be a positive form (or what is the same thing,  $a$  and  $a\beta - \eta^2$  must be positive). If  $A', H', B'$  are also the parameters of a theta-function, then writing

$$A', H', B' = A'_0 + ia', H'_0 + i\eta', B'_0 + i\beta',$$

$(a', \eta', \beta' \mid x, y)^2$  must also be a positive form. It can be shown that,  $A', H', B'$  being determined as above, the former condition implies the latter one; viz. if the form  $(a, \eta, \beta \mid x, y)^2$  be positive, then also the form  $(a', \eta', \beta' \mid x, y)^2$  will be positive.

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Write

$$\begin{aligned} A &= A_0 + ia, & A' &= A'_0 + ia', \\ B &= B_0 + i\beta, & B' &= B'_0 + i\beta', \\ H &= H_0 + i\eta, & H' &= H'_0 + i\eta', \\ \Delta &= \Delta_0 + i\delta, \\ \Delta_0 &= H_0^2 - A_0B_0 - (\eta^2 - a\beta), \\ \delta &= 2H_0\eta - B_0a - A_0\beta, \end{aligned}$$

and let the quintic matrix in the foregoing transformation

$$(T', P', Q', R', S') = (M)(T, P, Q, R, S)$$

be represented by

$$\begin{pmatrix} \cdot & t & p & q & r & s \\ 1 & & & & & \\ 2 & & & & & \\ 3 & & & & & \\ 4 & & & & & \end{pmatrix}.$$

**43.** It is to be shown that  $a'x^2 + 2\eta'xy + \beta'y^2$  is definite and positive.

We require  $a', \beta', \eta'$ ; we have

$$A' + ia' = \frac{(t_1, p_1, q_1, r_1, s_1)(1, A_0 + ia, H_0 + i\eta, B_0 + i\beta, \Delta_0 + i\delta)}{(t_0, p_0, q_0, r_0, s_0) \cdots} = \frac{M_1 + N_1i}{M_0 + N_0i},$$

and thence

$$a' = \frac{M_0N_1 - M_1N_0}{M_0^2 + N_0^2} = \frac{1}{\square}(M_0N_1 - M_1N_0).$$

Now

$$\begin{aligned} M_0N_1 - M_1N_0 &= (t_0p_1 - t_1p_0)a \\ &\quad + (t_0q_1 - t_1q_0)\eta \\ &\quad + (t_0r_1 - t_1r_0)\beta \\ &\quad + (t_0s_1 - t_1s_0)\delta \\ &\quad + (p_0q_1 - p_1q_0)(A_0\eta - H_0a) \\ &\quad + (p_0r_1 - p_1r_0)(A_0\beta - B_0a) \\ &\quad + (p_0s_1 - p_1s_0)(A_0\delta - \Delta_0a) \\ &\quad + (q_0r_1 - q_1r_0)(H_0\beta - B_0\eta) \\ &\quad + (q_0s_1 - q_1s_0)(H_0\delta - \Delta_0\eta) \\ &\quad + (r_0s_1 - r_1s_0)(B_0\delta - \Delta_0\beta); \end{aligned}$$

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and so for  $\eta', \beta'$  with only the change of  $t_1, p_1, q_1, r_1, s_1$  into  $t_2, p_2, q_2, r_2, s_2$ , and  $t_3, p_3, q_3, r_3, s_3$  respectively; we find, omitting a factor  $\frac{k}{\square}$  throughout,

|                             | $a' =$               | $\eta' =$                                         | $\beta' =$           |
|-----------------------------|----------------------|---------------------------------------------------|----------------------|
| $a$                         | $b_1^2$              | $-a_1b_1$                                         | $a_1^2$              |
| $\eta$                      | $-2b_1b_3$           | $a_1b_3 + a_3b_1$                                 | $-2a_1a_3$           |
| $\beta$                     | $b_3^2$              | $-a_3b_3$                                         | $a_3^2$              |
| $\delta$                    | $-(b_1b_2 + b_0b_3)$ | $\frac{1}{2}(a_1b_2 + a_2b_1 + a_0b_3 + a_3b_0)$  | $-(a_1a_2 + a_0a_3)$ |
| $A_0\eta - H_0a$            | $-2b_1b_2$           | $a_1b_2 + a_2b_1$                                 | $-2a_1a_2$           |
| $H_0\beta - B_0\eta$        | $-2b_2b_3$           | $a_2b_3 + a_3b_2$                                 | $-2a_2a_3$           |
| $A_0\beta - B_0a$           | $b_0b_3 - b_1b_2$    | $-\frac{1}{2}(a_1b_2 + a_2b_1 - a_0b_3 - a_3b_0)$ | $a_0a_3 - a_1a_2$    |
| $A_0\delta - \Delta_0a$     | $-b_1^2$             | $a_1b_1$                                          | $-a_1^2$             |
| $H_0\delta - \Delta_0\eta$  | $-2b_1b_2$           | $a_1b_2 + a_2b_1$                                 | $-2a_1a_2$           |
| $B_0\delta - \Delta_0\beta$ | $-b_3^2$             | $a_3b_3$                                          | $-a_3^2$             |

and hence

$$a'x^2 + 2\eta'xy + \beta'y^2 \\ = \frac{k}{\square} \{ \beta, -\eta, -\eta H_0 + \beta A_0, -\eta B_0 + \beta H_0 \mid x, y \}^2.$$

The right-hand side is here definite and positive (*suprà* No. 28), hence also the left-hand is definite and positive.

44. As a specimen of the work, observe that we have

$$\begin{aligned} t_0p_1 - t_1p_0 &= (a_0c_1 - a_1c_0)(ab)_{01} \\ &= b_1\{b_1(ac)_{01} + b_0(ac)_{10} + b_1(ac)_{01}\} \\ &= b_1\{-b_1(bd)_{01} - b_0(bd)_{10} + b_1\{-(bd)_{01} + k\}\} \\ &= b_1^2 \cdot k, \end{aligned}$$

since the remaining terms destroy each other. Dividing by  $\square$ , and then omitting the factor  $\frac{k}{\square}$ , we have thus the term  $ab_1^2$  in the foregoing expression for  $a'$ .

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45. Consider  $(a, b, c, d)_{0,1,2,3}$ , the components of a quartic matrix as above, and also  $T, P, Q, R, S; T', P', Q', R', S'$  connected as already mentioned; and write for shortness

$$\begin{aligned} z_0 &= a_0x + b_0y, & z_1 &= a_1x + b_1y, \\ z_2 &= a_2x + b_2y, & z_3 &= a_3x + b_3y, \end{aligned}$$